Solution to Laplace & Poisson Equations

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Aman Singh

CERTIFICATE

This is to certify that the project work entitled Solution to Laplace and Poisson Equations shows a work done by Aman Singh, for Course Computational Physics (PHB 61AL) at Jamia Millia Islamia, New Delhi, under my supervision and guidance.

I hereby approve it for submission to the university for the Course Project. Further, to the best of my knowledge, this report has not been submitted to any other institution for the award of any degree or diploma.

Dr. Syed Rashid Ahmad

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Introduction

The primary task of electrostatics is to find the electric field of a given stationary charge distribution. In theory, this purpose is accomplished by Coulomb's Law, given by

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\varepsilon}\int \frac{\vec{r}}{r^3}\ \rho(\vec{r}')\ d\tau'$$

The best strategy though is to calculate potential, V, given by

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon}\int \frac{1}{r}\ \rho(\vec{r}')\ d\tau'$$

This kind of integral can become complex to solve so we usually resort to the differential form of the Equation given by

$$\nabla^2 V = -\frac{1}{\varepsilon}\rho$$

This equation is known as Poisson's Equation. With appropriate boundary condition, this equation gives a unique value for potential for the given problem. When the charge density is 0, we have Laplace's Equation given by

$$\nabla^2 V = 0$$

∇^2 *is known as laplace operator, It act as a space derivative, and acts on a scalar function*

The above Laplace equation can be written in Cartesian Coordinate as

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

The Electric field can by calculated by taken by taking the taking the gradient of V calculated from the above method, E is given by

$$\vec{E} = -\nabla V$$

The gradient operator takes derivative of V with respect to x, y, z and coverts scalar into a vector,

$$\vec{E} = -\nabla V = -\frac{\partial V}{\partial x}\vec{i} - \frac{\partial V}{\partial y}\vec{j} - \frac{\partial V}{\partial z}\vec{k}$$

Theory

Laplace's Equation is a kind of averaging instruction; it tells you to assign to the point the average value of the surrounding value. Solution to Laplace equation in a sense is a averaging instruction, and yet it fits the point perfectly.

Laplace Equation tolerates no local maxima aur minima; extreme values of V must occur at the end points

Let us consider a two-dimensional case for solving the equation

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

The value of a point will be equal to the value of all the points surrounding it i.e. we the average value of the circle is equal to the value at the centre of the circle.

In order to solve this problem computationally we use method of relaxation, starting with a specified value of V at the boundary and a reasonable guess for V in the interior points, the first pass reassign to each point the average value of its nearest neighbour. The second pass repeats the process using previous situation and so on. After few iterations the number starts to settle down and a solution to the equation is achieved.

For Poisson's Equation the 0 is replaced with charge density/epsilon.

There are many ways to solve the Laplace's Equation computationally one such method known as finite difference method (Jacobi's Method) with successive relaxation method is used to calculate the solution of Laplace's Equation.

Computational Method Used to find the solution

From a numerical perspective the Laplace Equation is quite different from other derivatives.

Here we have Three independent variable instead of one, in which case we cannot use Euler RK4 or any other method. Laplace Equation is a partial derivative equation, there is no general-purpose algorithm to sole PDE. The solution depends upon the class of PDE (elliptical, parabolic, hyperbolic). Laplace equation belong to elliptical class. In order to solve it we do following method. First, we rewrite the equation as

$$\frac{\partial V}{\partial x} \approx \frac{V(i+1,j,k) - V(i-1,j,k)}{2\,\Delta x}$$

This can be used to further calculate the second derivative

$$\frac{\partial^2 V}{\partial x^2} \approx \frac{1}{\Delta x}\left[\frac{V(i+1,j,k) - V(i,j,k)}{\Delta x} - \frac{V(i,j,k) - V(i-1,j,k)}{\Delta x}\right]$$

On further rearranging,

$$\frac{\partial^2 V}{\partial x^2} \approx \frac{V(i+1,j,k) + V(i-1,j,k) - 2\,V(i,j,k)}{(\Delta x)^2}$$

Similarly taking the derivative for y and rearranging it further we get

$$V(i,j) = \frac{1}{4}\left[V(i+1,j,k) + V(i-1,j,k) + V(i,j+1,k) + V(i,j-1,k)\right]$$

Here we implement the Jacobi method, The core of such problem must repeatedly update V(i,j) until it reaches convergence. So we start by giving boundary condition and an guess for the internal value of V(i,j). The program starts to calculate values for the points in the grid and saves the data then it checks for convergence, if the convergence is not meet the program recalls the earlier value and starts to calculate values by taking the average of the surrounding points and again checks for convergence. This process keeps on continuing until the convergence is reached. In order to check convergence, the cumulative change is used to judge convergence.

Finally, if we wish to calculate the electric field, this can be calculated by differentiating V

$$E_x = -\frac{\partial V}{\partial x}$$

With corresponding relations for Ey. This derivative can be estimated using our usual finite difference method,

$$E_x(i,j) \approx -\frac{V(i+1,j) - V(i-1,j)}{2\,\Delta x}$$

Similar Expression can be used to calculate Ey.

In case of Poisson's Equation, we include the charge density,

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho}{\epsilon_0}$$

$$V_2(i,j,k) = \frac{1}{6}[V(i-1,j,k) + V(i+1,j,k) + V(i,j+1,k) + V(i,j-1,k) + V(i,j,k+1) + V(i,j,k-1)] + \frac{\rho(i,j,k)\Delta x^2}{6},$$

the one written here is for three dimension which can be changed for 2D.

In the case of Poisson's Equation, the value of potential is average of its surrounding + the potential due to the local charge distribution.

Computer code using Fortran90

```
! This is a program to solve Laplace and Poisson equations
! We use finite difference method to solve the equations

  program solution
  implicit none

  ! Define parameter
  integer, parameter  :: nx = 100, ny = 100 ! Dimension of grid
  real, parameter :: dx = 0.1, dy = 0.1
  real, parameter :: epsilon0 = 8.854e-12 ! Permittivity of free space
  real, parameter :: eps = 1.0e-6 ! Convergence condition

  ! Define potential, charge density, Electric Field
  real, dimension(nx,ny) :: phi_laplace, phi_poisson
  real, dimension(nx,ny) :: rho
  real, dimension(nx,ny) :: Ex_laplace, Ey_laplace
  real, dimension(nx,ny) :: Ex_poisson, Ey_poisson


  ! Define variable
  integer :: i, j, iter
  real :: max_er, er

  ! Initial conditions for potentials
  phi_laplace = 0.0
  phi_poisson = 0.0
```

```
! Initial conditions for charge density
rho = 0.0

! Initial condition of Electric Field
Ex_laplace = 0.0
Ey_laplace = 0.0
Ex_poisson = 0.0
Ey_poisson = 0.0

! Boundary condition for potentials
phi_laplace(1,:) = 0.0
phi_laplace(nx,:) = 0.0
phi_laplace(:,1) = 0.0
phi_laplace(:,ny) = 0.0

phi_poisson(1,:) = 0.0
phi_poisson(nx,:) = 0.0
phi_poisson(:,1) = 0.0
phi_poisson(:,ny) = 0.0

! Set the charge density
rho(nx/4,ny/4) = 1.0 * epsilon0
rho(3*nx/4,3*ny/4) = 1.0 * epsilon0
rho(3*nx/4,ny/4) = -1.0 * epsilon0
rho(nx/4,3*ny/4) = -1.0 * epsilon0



! Solve Laplace equation
```

```
iter = 0
max_er = 1.0
do while (max_er > eps .and. iter < 10000)
max_er = 0.0
 do i = 2, nx-1
  do j = 2, ny-1
  phi_laplace(i,j) = 0.25*(phi_laplace(i+1,j) + phi_laplace(i-1,j) &
                    + phi_laplace(i,j+1) + phi_laplace(i,j-1))
  er = abs(phi_laplace(i,j)-phi_laplace(i-1,j))
  if (er > max_er) max_er = er
  enddo
 enddo
 iter = iter + 1
enddo

! Solve Poisson equation
iter = 0
max_er = 1.0
do while (max_er > eps .and. iter < 10000)
max_er = 0.0
 do i = 2, nx-1
  do j = 2, ny-1
  phi_poisson(i,j) = 0.25*(phi_poisson(i+1,j) + phi_poisson(i-1,j) &
              + phi_poisson(i,j+1) + phi_poisson(i,j-1)) &
              + 0.25*dx*dy*rho(i,j)/epsilon0
  er = abs(phi_poisson(i,j)-phi_poisson(i-1,j))
  if (er > max_er) max_er = er
  enddo
 enddo
```

```
 iter = iter + 1
enddo

! Calculate the electric field for laplace equation
do i = 2, nx-1
 do j = 2, ny-1
  Ex_laplace(i,j) = -0.5*(phi_laplace(i+1,j)-phi_laplace(i-1,j))/dx
 enddo
enddo

do j = 2, nx-1
 do i = 2, ny-1
  Ey_laplace(i,j) = -0.5*(phi_laplace(i,j+1)-phi_laplace(i,j-1))/dy
 enddo
enddo

! Calculate the electric field for poisson equation
do i = 2, nx-1
 do j = 2, ny-1
  Ex_poisson(i,j) = -0.5*(phi_poisson(i+1,j)-phi_poisson(i-1,j))/dx
 enddo
enddo

do j = 2, nx-1
 do i = 2, ny-1
  Ey_poisson(i,j) = -0.5*(phi_poisson(i,j+1)-phi_poisson(i,j-1))/dy
 enddo
enddo
```

```
! Write the potential and the electric field to a file for laplace
open(unit=1, file="potentiallaplace.txt", status="replace")
 do i = 1, nx
  do j = 1, ny
   write(1,*) i*dx, j*dy, phi_laplace(i,j)
  enddo
enddo
close(1)

open(unit=2, file="efieldlaplace.txt", status="replace")
do i = 1, nx
 do j = 1, ny
  write(2,*) i*dx, Ex_laplace(i,j), j*dy, Ey_laplace(i,j)
 enddo
enddo
close(2)

! Write the potential and the electric field to a file for poisson
open(unit=3, file="potentialpoisson.txt", status="replace")
do i = 1, nx
 do j = 1, ny
  write(3,*) i*dx, j*dy, phi_poisson(i,j)
 end do
end do
close(3)

open(unit=4, file="efieldpoisson.txt", status="replace")
do i = 1, nx
 do j = 1, ny
```

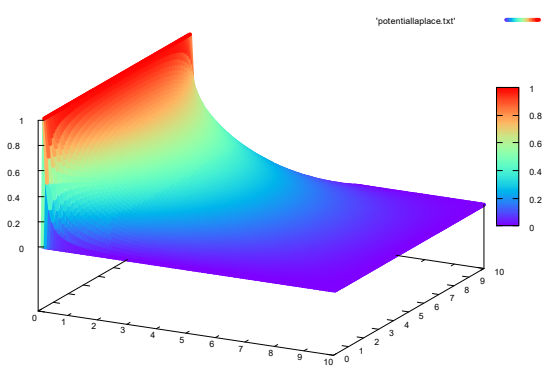
```
  write(4,*) i*dx, Ex_poisson(i,j), j*dy, Ey_poisson(i,j)
 end do
end do
close(4)

end program solution

! This is a program to solve Laplace Special Case
! We use finite difference method to solve the equations

program solution
implicit none

! Define parameter
integer, parameter  :: nx = 100, ny = 100 ! Dimensions of grid
real, parameter :: dx = 0.1, dy = 0.1
real, parameter :: epsilon0 = 8.854e-12 ! Permittivity of free space
real, parameter :: eps = 1.0e-6 ! Convergence condition

! Define potential, Electric Field
real, dimension(nx,ny) :: phi_laplace
real, dimension(nx,ny) :: Ex_laplace, Ey_laplace

! Define variable
integer :: i, j, iter
real :: max_er, er

! Initial conditions for potentials
phi_laplace = 0.0
```

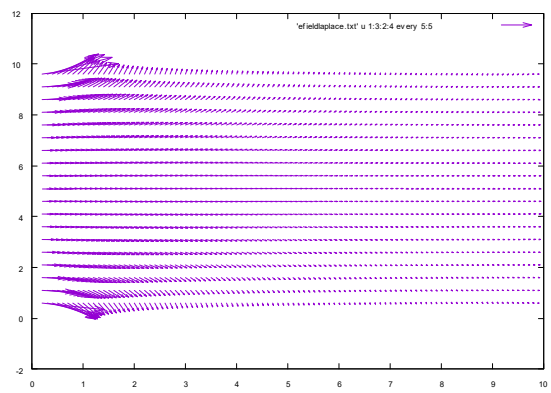
```
! Initial condition of Electric Field
Ex_laplace = 0.0
Ey_laplace = 0.0

! Boundary condition for potentials
phi_laplace(1,:) = 0.0
phi_laplace(nx,:) = 0.0
phi_laplace(:,1) = 0.0
phi_laplace(:,ny) = 0.0


! Solve Laplace equation
iter = 0
max_er = 1.0
do while (max_er > eps .and. iter < 10000)
max_er = 0.0
 do i = 2, nx-1
 if ( i >= nx/4 .and. i<= 3*nx/4) then
 do j = ny/4, 3*ny/4
  phi_laplace(i,j) = 1.0
 enddo
 endif
  do j = 2, ny-1
  phi_laplace(i,j) = 0.25*(phi_laplace(i+1,j) + phi_laplace(i-1,j) &
                    + phi_laplace(i,j+1) + phi_laplace(i,j-1))
  er = abs(phi_laplace(i,j)-phi_laplace(i-1,j))
  if (er > max_er) max_er = er
  enddo
```

```
 enddo
 iter = iter + 1
enddo

! Calculate the electric field for Laplace equation
do i = 2, nx-1
 do j = 2, ny-1
  Ex_laplace(i,j) = -0.5*(phi_laplace(i+1,j)-phi_laplace(i-1,j))/dx
 end do
end do

do j = 2, nx-1
 do i = 2, ny-1
  Ey_laplace(i,j) = -0.5*(phi_laplace(i,j+1)-phi_laplace(i,j-1))/dy
 end do
end do


! Write the potential and the electric field to a file for Laplace
open(unit=5, file="potentiallaplace2d.txt", status="replace")
 do i = 1, nx
  do j = 1, ny
   write(5,*) i*dx, j*dy, phi_laplace(i,j)
  end do
end do
close(5)

open(unit=6, file="efieldlaplace2d.txt", status="replace")
do i = 1, nx
```

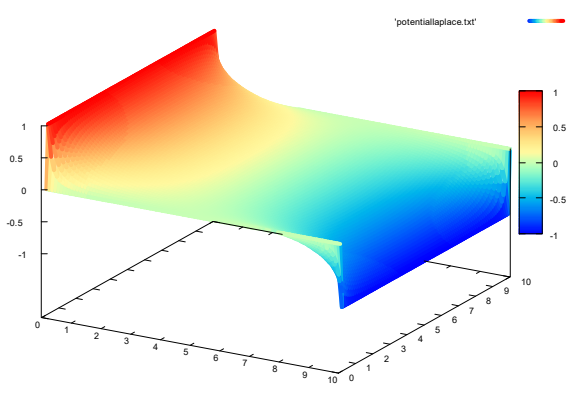
```
 do j = 1, ny
  write(6,*) i*dx, Ex_laplace(i,j), j*dy, Ey_laplace(i,j)
 end do
end do
close(6)


end program solution
```

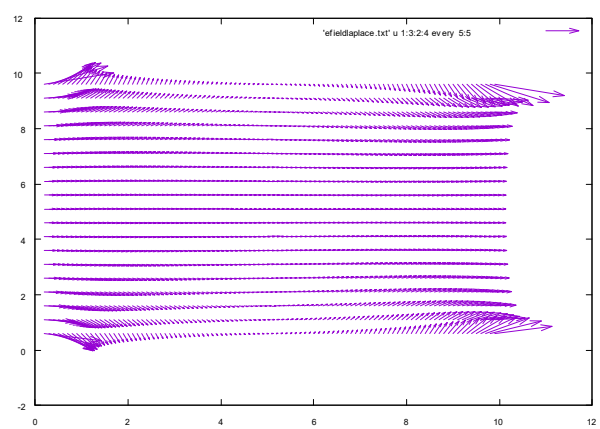
Outputs



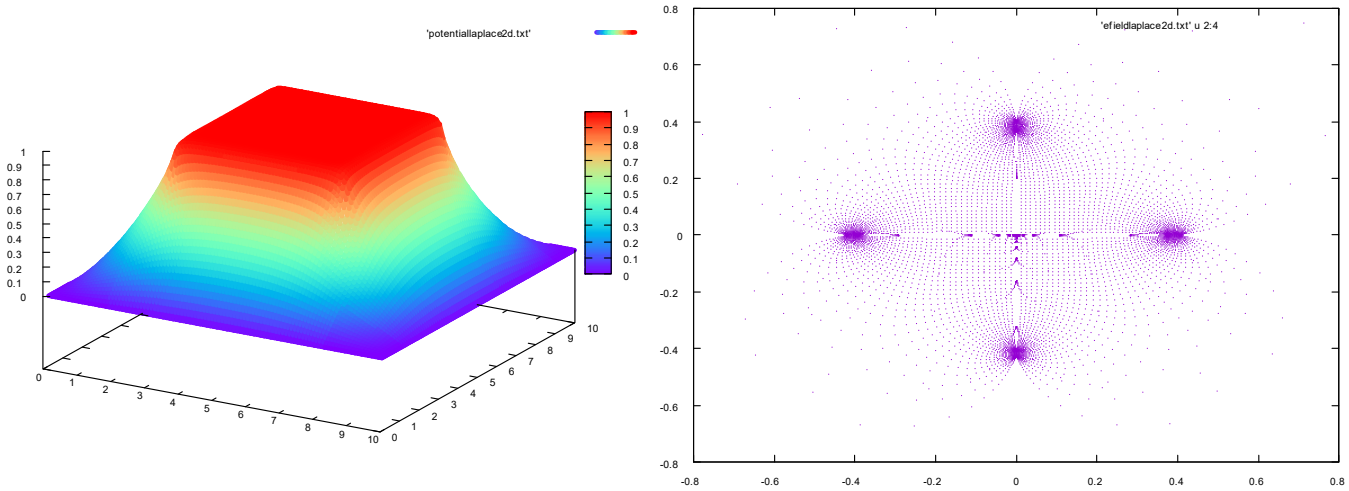


One side of the box is held at +1 V while other is at 0 V

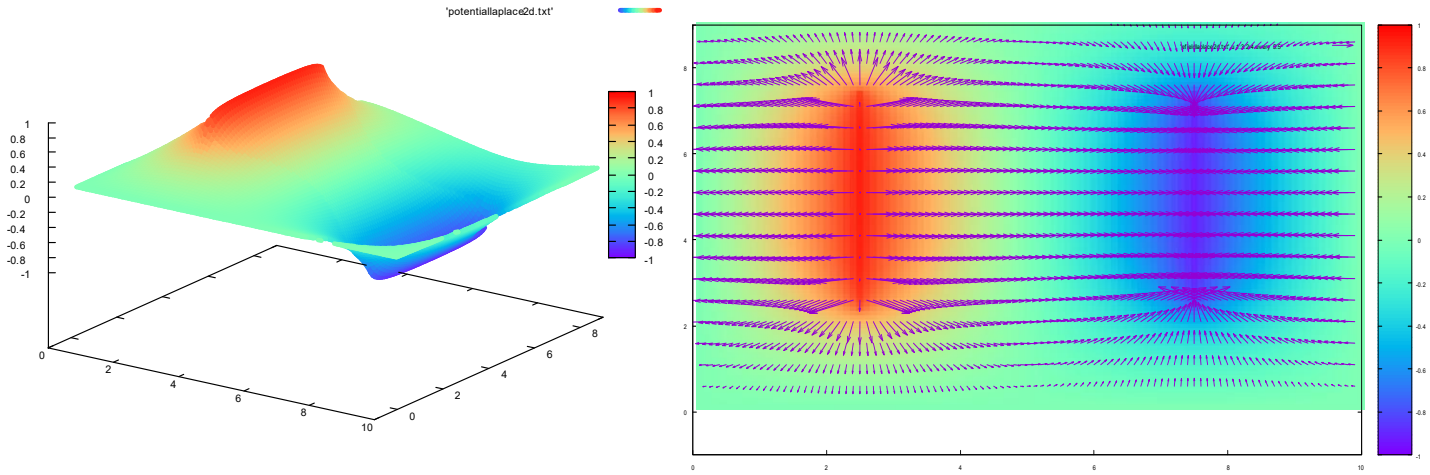




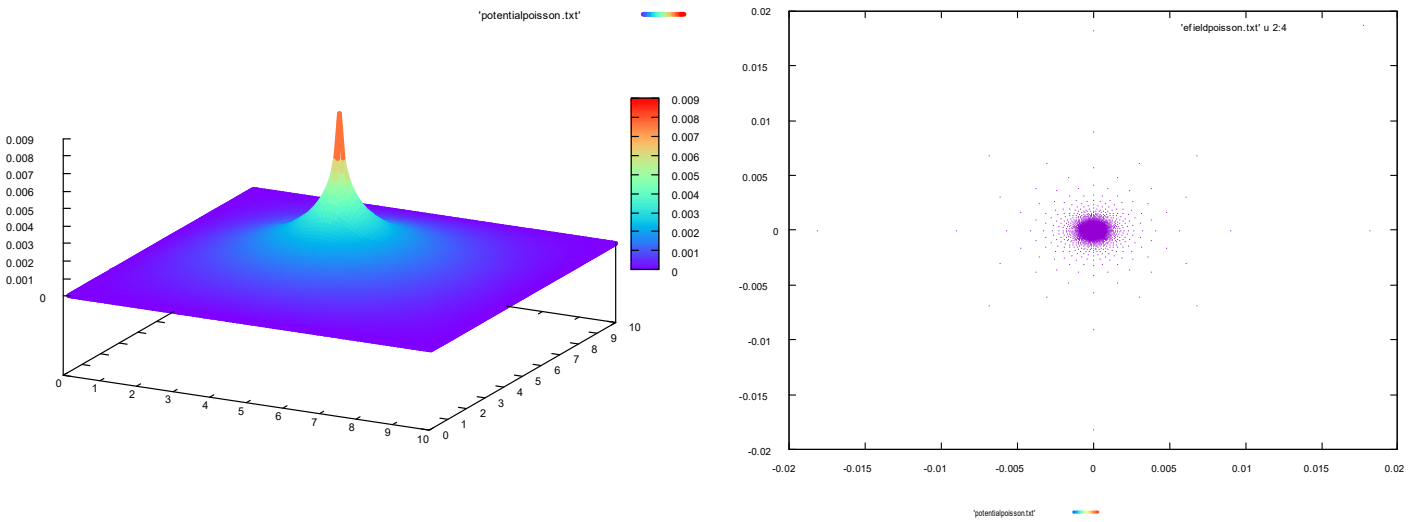
One side is held at +1 V while other is at -1 V



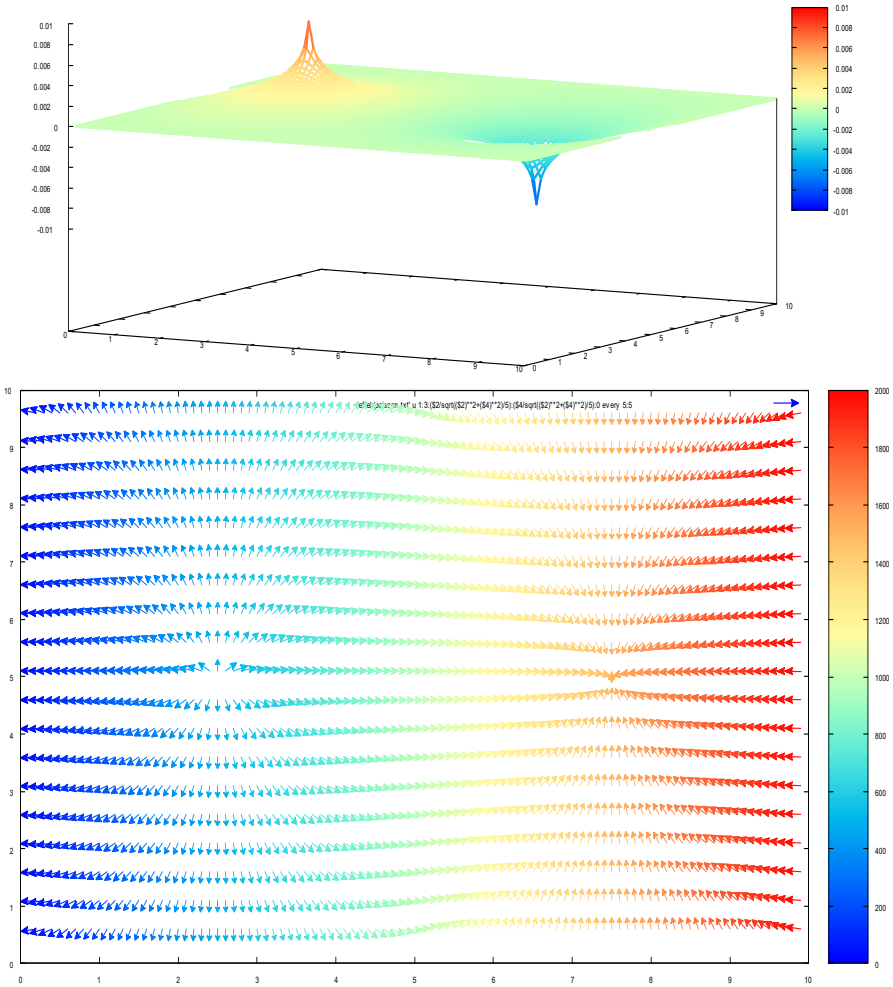
A 2D Bar kept inside the 2D box



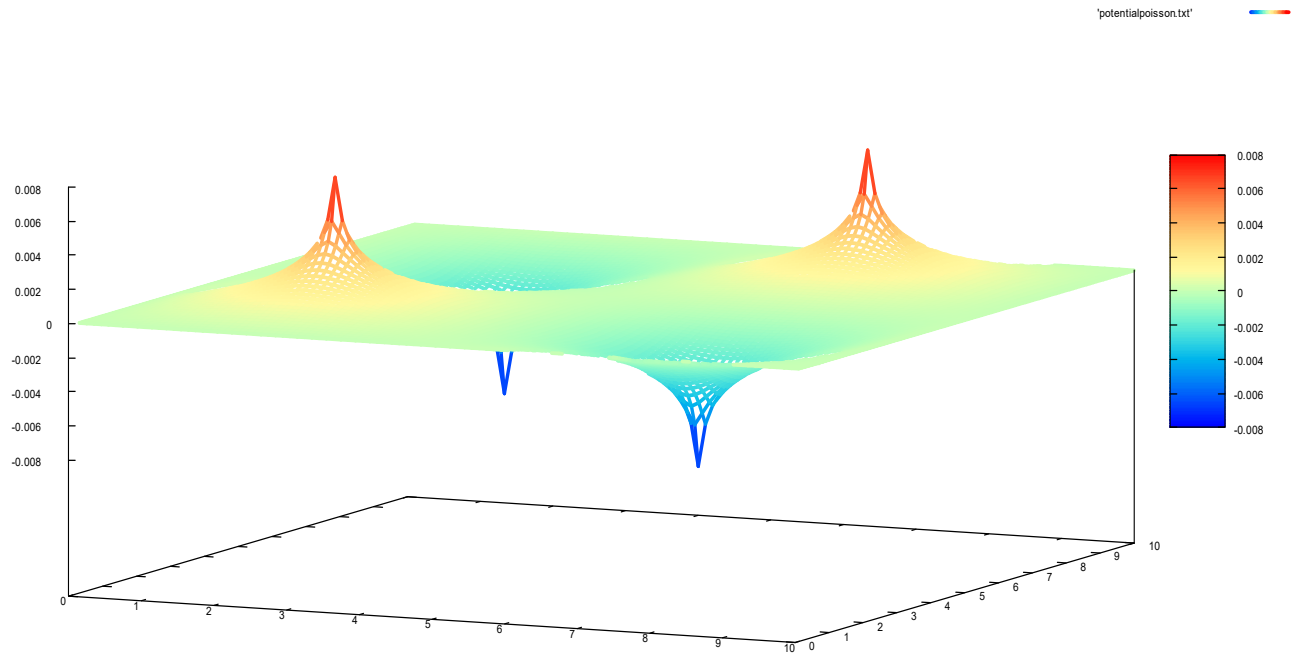
Conductor Plates kept inside the box maintained at +1, -1 v

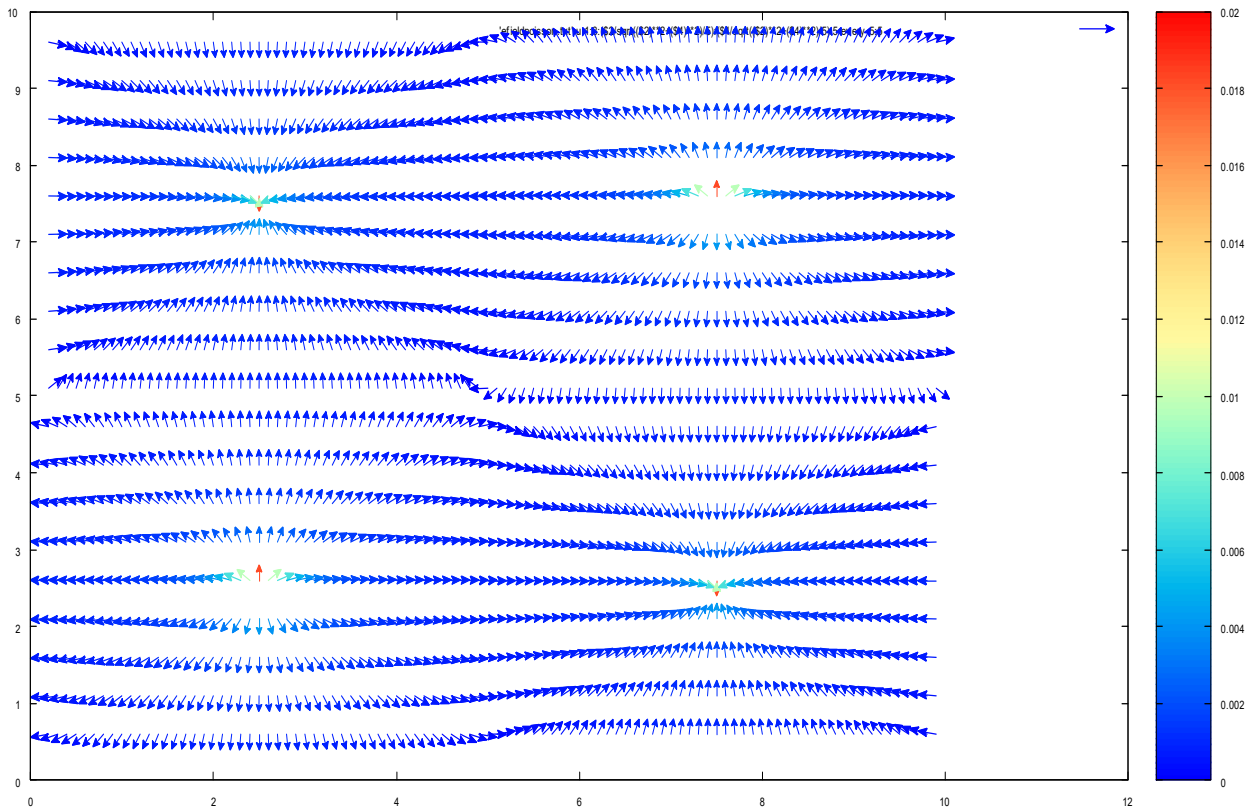


Monopole inside a 2D box



Dipole inside a 2D box





Quadrupole kept inside a 2D box

Conclusions

As we can see the following graphs show potential and Electric Field in Different case. In case of charge free space, we follow Laplace's equation. The graphs shows that the potential is continuous and smooth. There is no local maxima aur minima, extreme values occur only at extreme points. The electric field lines do not seem to converge or diverge from a point in space.

While in case where charge is present, we see sudden peak in potential values which fall with R as we move away from the charge. We also see that where charge is present the Electric field seem to diverge (in case of +ve charge), and diverge (in case of -ve) charge.

The potential or Electric field graph we see here are rather different from what we see in case of Theoretical approach. The discrepancy arises from the fact that in case of theory we take 0 of potential to be at infinity. Whereas in practical or the following case where we have taken the 0 of potential to be the box gives different value for potential and Electric Field.

Also, in case of Dipole and Quadrupole we can clearly see that the Electric field starts from +ve charge and ends at -ve charge. The box is considered to be metallic kept at 0 V, the field abruptly disappear at the box.

The case here we studied is of Electrostatic, In Dynamic case we also have Magnetic field and accelerating charges gives rise to radiations. The Laplace and Poisson Equation in case of Electrodynamics also have a second order time derivative, which gives solution for Electric and magnetic field changing with time.